

Let's revisit our exercise:

$\langle S \rangle$ i saw a cat ✓
 $\langle S \rangle$ i saw a dog
 $\langle S \rangle$ i saw the dog
 $\langle S \rangle$ i jumped
 $\langle S \rangle$ a cat jumped high
 $\langle S \rangle$ the dog barked loud
 $\langle S \rangle$ a cat jumped high
 $\langle S \rangle$ the cat slept

Assuming a bigram model, compute:

1. $p(\text{the cat jumped})$ ←
2. $p(\text{the cat barked})$
3. $p(\text{the dog barked})$
4. $p(\text{i saw the dog})$

Vocabs = $\langle S \rangle$ jumped
 i
 saw high
 a barked
 cat loud
 the slept
 dog $\langle S \rangle$

=

$$P(\langle S \rangle \text{ the cat barked})$$

$$= P(\text{the} | \langle S \rangle) \times P(\text{cat} | \text{the}) \times P(\text{barked} | \text{cat})$$

$$= \frac{2+1}{8+13} \times \frac{1+1}{3+13} \times \frac{0+1}{4+13}$$

$$= \frac{3}{21} \times \frac{2}{16} \times \frac{1}{17}$$

$$= ?$$

$$P(\langle S \rangle \text{ the cat jumped})$$

$$= P(\text{the} | \langle S \rangle) \times P(\text{cat} | \text{the}) \times P(\text{jumped} | \text{cat})$$

$$= \frac{3}{21} \times \frac{2}{16} \times \frac{2+1}{4+13} \times \frac{3}{17}$$

$$= \frac{1}{7} \times \frac{1}{8} \times \frac{3}{17} =$$